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Effectiveness of physiotherapy modalities on persisting dyspnoea in long COVID: A systematic review and meta-analysis

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ABSTRACT

Background: Dyspnoea is often found months and years later in the "long-covid" syndrome, impairing quality of life and further perpetuating anxiety and post-traumatic stress disorders. Physiotherapy was recommended as a treatment in long-covid, but there is still insufficient evidence on its effectiveness.

Methods: We conducted a systematic literature search on MEDLINE, PEDro, WOS, Scopus, VHL and the Cochrane Library until July 2023 (PROSPERO registration number: CRD42023427464). We selected comparative trials including adults with persistent breathlessness following COVID-19, regardless of the initial severity, for whom physiotherapy was implemented as a treatment for dyspnoea. We followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and assessed the study quality using the PEDro Scale.

Results: 19 studies that included 1292 adults fulfilled the inclusion criteria, of which 15 were randomised controlled trials and 4 non-randomised controlled trials. As for the rehabilitation modalities, 6 studies used respiratory muscle training, 6 studies used low to moderate intensity rehabilitation, 6 used high intensity rehabilitation and one used passive rehabilitation. The methods used between and within each group differed greatly, leading to an expected high heterogeneity of results. Nethertheless the random-effects model found a significant difference favouring physiotherapy (SMD -0.63, 95 CI [-1.03; −0.24], $p < 0.001$, $I^2 = 88$ %). Subgroup analysis showed a significant effect in the high intensity rehabilitation group alone, with null heterogeneity.

Conclusion: In people suffering from dyspnoea following a SARS-CoV-2 infection, physiotherapy and especially pulmonary rehabilitation may help alleviate respiratory symptoms. Future studies will need to provide more consistent rehabilitation methods and better descriptions of them so as to reveal clear effects and avoid the confusion caused by using too many rehabilitation modalities.

1. Introduction

The COVID-19 pandemic has placed the world's healthcare systems under considerable pressure, leading to an upsurge in the number of people admitted to hospitals with acute respiratory failure. Although most SARS-CoV-2 infections resolved naturally, 10 % of hospitalised people developed a hyperinflammatory state [1] leading to an alteration in lung tissue properties and profound hypoxemia [2]. The pulmonary parenchyma then underwent a loss of structural integrity caused by the progression of inflammatory lesions, potentially leading to diffuse alveolar damage or fibrosis. This alveolo-interstitial edema significantly reduces both pulmonary tissue compliance and alveolar ventilation, ultimately increasing respiratory workload. These phenomena contribute to the heterogeneity of the ventilation/perfusion ratio within the lung, resulting in the emergence of often severe hypoxemia and mechanical stress (patient self-inflicted lung injuries) on the parenchyma, exacerbating local and systemic inflammation [3].

However, the viral infection not only induces inflammatory lesions within the lung parenchyma but also disrupts other organs, such as skeletal muscles [3–8]. With the continuing progression of those lung injuries, an alteration of respiratory muscles may develop, triggered by

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Abbreviations:

ARDS	Acute Respiratory Distress Syndrome
CARDS	COVID-19 Related Acute Respiratory Distress Syndrome
COPD	Chronic Obstructive Pulmonary Disease
EMT	Expiratory Muscle Training
HRQOL	Health-Related Quality Of Life
ICU	Intensive Care Unit
IMT	Inspiratory Muscle Training
LLLT	Low-Level Laser Therapy
MCID	Minimal Clinically Important Difference
MDP	Multidimensional Dyspnea Profile
MeSH	Medical Subject Headings
mMRC	modified Medical Research Council
PR	Pulmonary Rehabilitation
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PTSD	Post Traumatic Stress Disorder
RCT	Randomised Controlled Trial
RMT	Respiratory Muscle Training

the combination of inflammation and exhaustion in response to increased work of breathing and hypoxia. Respiratory acidosis and muscle fatigue, particularly affecting the diaphragm, eventually become a pivotal factor in the manifestation of dyspnoea [9]. Consequently, individuals affected by COVID-19 often experience prolonged periods of bed rest during and post-hospitalisation [10], creating a favourable environment for the development of peripheral amyotrophy and continuing inflammatory state. The resulting peripheral amyotrophy, characterised by the wasting of skeletal muscles, not only amplifies the loss of autonomy but also contributes to the chronicity of symptoms [3,4,6]. In those surviving the initial condition, this intricate interplay of inflammatory state, muscle fatigue, and peripheral amyotrophy sets the stage for long-term consequences of COVID-19 on respiratory health and overall individual well-being [6]. As a matter of fact, up to 15 % of people hospitalised for COVID-19 eventually developed a "long-covid", roughly similar to the post-intensive care syndrome [11–13], despite the initial rehabilitation in intensive care units. Long-covid is defined as "the continuation or development of new symptoms 3 months after the initial SARS-CoV-2 infection, regardless of initial severity" [13], and characterised by numerous person-dependent symptoms impacting different systems (neurologic, digestive, musculoskeletal and cardiovascular) whose severity may vary from mild to incapacitating. With fatigue, muscle weakness and anxiety, dyspnoea is one of the major symptoms reported in 40–50 % of people suffering with "long-covid" and even more so following the severe form of COVID-19 [6,14,15].

Dyspnoea, just like pain, is a challenging symptom to treat [16]. This persistent feeling of shortness of breath is a strong source of fatigue, anxiety and even fear of dying and contributes to the occurrence of post-traumatic stress disorders (PTSD). Dyspnoea is indeed multifactorial (emotional, sensorial and functional components), unpleasant and always significant [17]. In the end, dyspnoea and its consequences on both physical and mental health and on the individual's autonomy in his or her activities of daily living, contributes greatly to reducing quality of life. For these reasons international recommendations stressed the importance of rehabilitation for those with long-covid to treat dyspnoea, fatigue and functional disability that persists after the initial infection, regardless of its original gravity [18–21].

Since COVID-19 was an emerging disease, most, if not all of these recommendations were modelled on chronic obstructive pulmonary disease (COPD) actual guidelines [20,22], comprising many rehabilitation modalities among which pulmonary rehabilitation (PR) holds the strongest evidence [23]. However, in recovering people with a healed or still healing lung, some degree of motor weakness and a wide panel of limbic responses to the dyspnoea, the specific acute physio-pathological mechanisms of COVID-19 may lead to a peculiar response to rehabilitation compared with COPD [24]. Furthermore, physiotherapy is a patient-centred care; it comprises by itself a wide range of treatments amongst which numerous and quite diverse techniques can be used to achieve a common therapeutic goal. Indeed, physiotherapy is tailor-made and dispatched through the hands of a physiotherapist; and individual responses to the treatment may vary because of the individuals' clinical presentations [25–27]. As a result there is no real categorisation of these techniques and their clinical appropriateness for one peculiar pathology is thus complex to measure. Rehabilitation can mainly be passive, without involving the individual (e.g. mobilisation, stretching, electrotherapy) or active (e.g. endurance and aerobic training, strength training, balance training); and in the latter case global or specific to a muscle group, at low or high intensity etc. Heterogeneity is well known in physiotherapy and the numerous modalities even prevent most meta-analyses to be made or published, rendering recommendations regarding their potential benefit a hard task, even for a seemingly simple condition like low-back-pain [28]. Moreover, due to the lack of studies and long-term experience regarding this new condition, international recommendations regarding rehabilitation in COVID and long-COVID were rather imprecise, allowing physiotherapists a certain amount of latitude in their treatment [19,20,29–31]. In the end, because of a lack of strong evidence-based recommendations, figuring out which techniques should be used in this specific long-covid application is a challenge in itself.

For these reasons we sought to assess the different treatments of physiotherapy and their specific effects on dyspnoea amongst people presenting persisting respiratory symptoms, following a Sars-CoV-2 infection, after being discharged from hospitals.

2. Methods

The study protocol was registered and published on June 14th, 2023 on the International Prospective Register of Systematic Reviews (PROSPERO) under the identification number CRD42023427464. Reporting of this systematic review and meta-analysis was made in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) [32].

2.1. Inclusion and exclusion criteria

Comparative clinical trials including adults with persistent rest or exertional breathlessness beyond hospitalisation for a SARS-CoV-2 infection, and for whom physiotherapy was implemented as treatment were eligible for inclusion. We included all peer-reviewed studies who had assessed breathlessness on a dyspnoea specific scale (e.g. Borg, Dyspnoea-12 etc.), a functional scale (e.g. Modified Medical Research Council dyspnoea scale) or integrated in a health-related quality of life scale (e.g. St George Respiratory Questionnaire). We excluded (i) protocol studies, (ii) studies in which dyspnoea was first assessed per-hospitalisation, (iii) studies in which dyspnoea was only evaluated indirectly through a functional test (e.g. spirometry, 6 min walking test).

Our search was conducted on MEDLINE (National Library of Medicine's premier bibliographic database), PEDro (Physiotherapy Evidence Database), WOS (Web of Science), Scopus, VHL (Virtual Health Library) and the Cochrane Library. Studies published in English, French, Spanish or Portuguese between September 1st, 2019 and July 2nd, 2023 were evaluated for inclusion.

2.2. Search strategy

In order to properly conduct the systematic review, we used a controlled combination of Medical Subject Headings (MeSH) terms and booleans operators covering concepts related to physiotherapy,

dyspnoea and COVID-19. As physiotherapy treatments are numerous and non-standardised in their labelling, a particularly inclusive combination available in the supplementary materials was used in order to avoid missing potential eligible studies.

As several databases were used with overlapping data, a semi-automation tool (Rayyan©) was used first to identify duplicates and ineligible articles (i.e. non-clinical trials). Two reviewers (CR, JW) independently screened titles and abstracts to select the articles meeting our inclusion criteria. The reviewers then independently reviewed all full texts assessed for eligibility and decided on inclusion. If necessary, the corresponding authors were contacted to retrieve incomplete data or to clarify methodology elements necessary for our study. Disagreements were resolved through discussion with a senior investigator (FP). A flow diagram of the research process is presented according to the PRISMA guidelines in Fig. 2.

2.3. Data collection and synthesis

Data were extracted in duplicate by the two reviewers using a standardised Excel® file; each reviewer independently collected data on the design, population, duration, outcomes. The following information was retrieved: registration number, date of publication, design, persons' inclusion criteria, demographic characteristics, interventions used and their durations, outcomes collected. In order to anticipate the major heterogeneity of rehabilitation interventions which could render any meta-analysis impossible (the different treatments proposed could potentially be as numerous as there were studies), we chose to sort interventions into 4 sub-groups following a logical physiotherapy algorithm for respiratory interventions (Fig. 1). Rehabilitation techniques with parameters specific to PR [22] (2–3 sessions per week over 45 min of training with a target between 4 and 6 on the borg scale) were classified into the High intensity group, while the other rehabilitation techniques were either classified in the Low-to-moderate intensity group when the durations and intensity were not those recommended for PR, or in the Respiratory muscle training group when the techniques were only specifically aimed at reinforcing respiratory muscles without whole-body rehabilitation. The last group "others" comprises techniques in which the person received passive treatment.

Finally, a physiotherapy clinical trial specific 10-items scale (PEDro scale) was used to assess the quality and risk of bias in included trials [33]. It comprises a score between 0 (high risk of bias) to 10 (low risk of bias). Any disagreements between the reviewers was resolved by involvement of the methodologist (GC).

2.4. Statistical analysis

As dyspnoea was measured on different scales, the interpretation of these criteria is challenging, as they are not always comparable with one another. To address this particular issue, standardised mean differences (SMD) were calculated to enable these different scales to be compared with each other, following the Cochrane Handbook for Meta-Analyses [34]. When one study scored dyspnoea on more than one scale, the most appropriate scale was chosen relative to rehabilitation (eg. modified Medical Research Council Dyspnoea (mMRC) scale rather than Borg). For studies reporting results as median, first and third quartiles, mean and standard deviation were estimated using Luo et al. and Wan et al. methods respectively [35,36]. For studies reporting binary outcomes, SMD were calculated using Murad's method [37]. A random-effect model was used to obtain the summary estimate of standardised mean differences. Data are presented using a forest plot (point estimates with their 95 % confidence intervals). Funnel plot was used to assess publication bias and influence analysis was performed by calculating pooled estimates omitting one study at a time. Heterogeneity was evaluated using forest plot inspection and the I^2 with a threshold value of more than 50 % for significant heterogeneity. A subgroup analysis according to intervention implemented was performed. This analysis was performed in the R language environment (version 4.2.2), using the metagen function from the General Package for Meta-Analysis

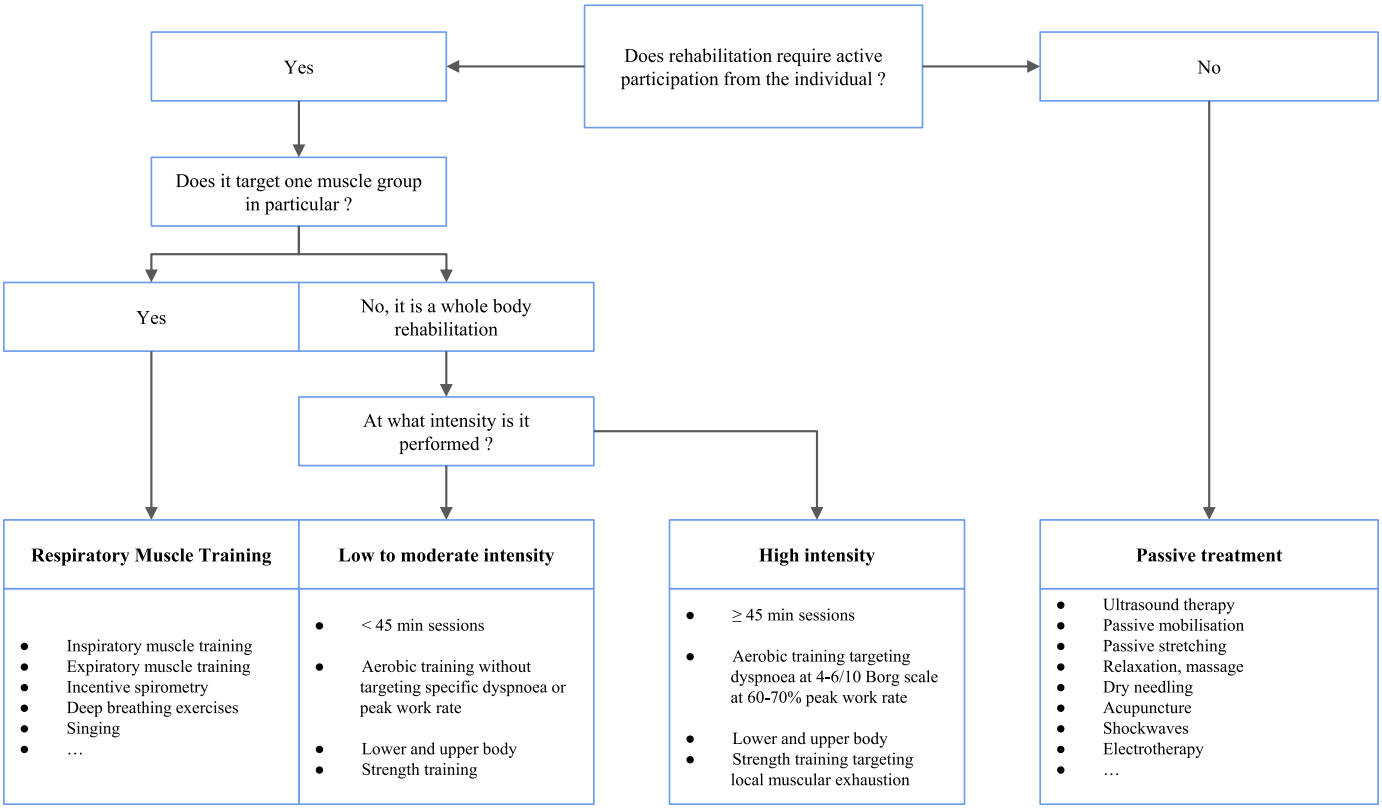

Fig. 1. Rehabilitation subgroup separation according to the intervention described.

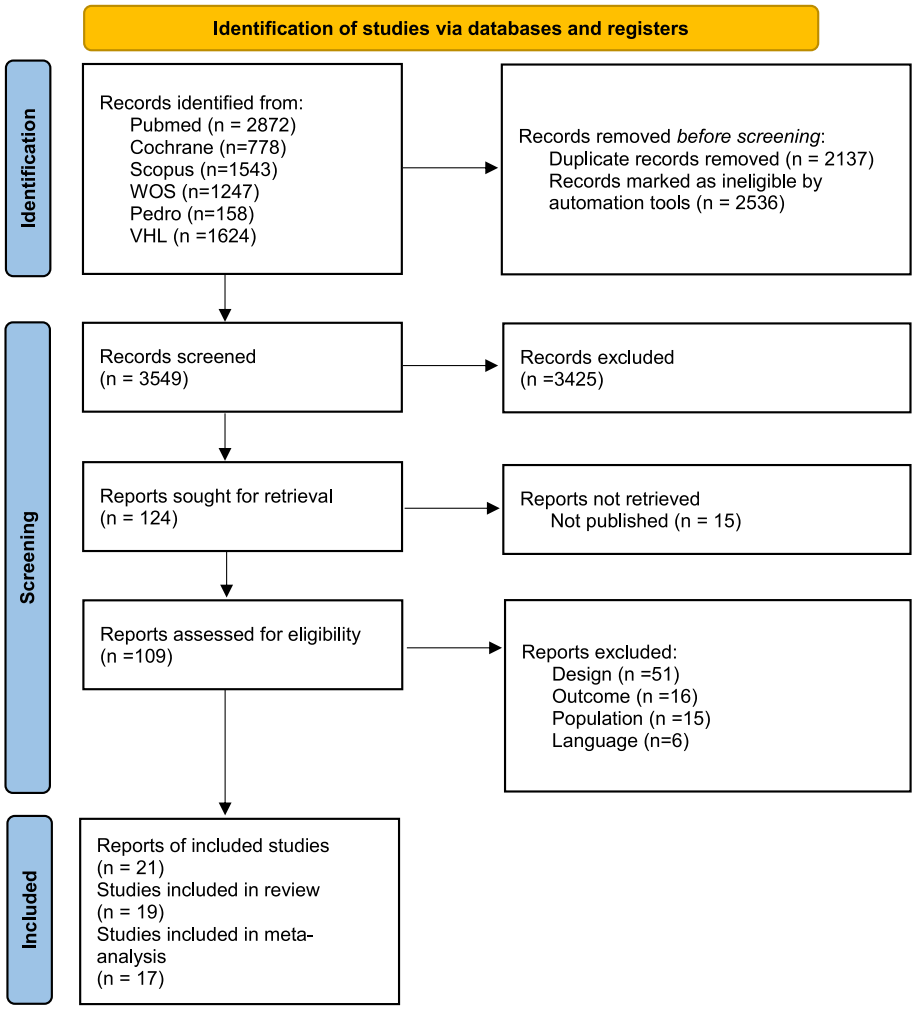

Fig. 2. 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram.

(meta v7.0.0) and the Meta-Analysis Package for R (metafor v4.4–0).

3. Results

The search strategy identified 8222 records (Fig. 2) as of July 2nd, 2023. After removing duplicates (2137), ineligible articles (2536) and publications that did not fit our inclusion criteria on the title and abstracts (3425), 124 articles remained, of which only 109 were published and retrieved for full text reading. Of these, 21 reports were finally selected. Four reports [38–41] shared a common registration number with a twin report. After verification with the authors, we chose to analyse only the latest iteration of both articles, leaving 19 articles for the systematic review, of which 15 were randomised controlled trials and 4 were non-randomised controlled trials. Finally two studies [41, 42] shared incomplete data and graphs without numbers thus preventing the use of data for a meta-analysis; they were hence excluded from the meta-analysis after failure to obtain said data.

Table 1 contains summarised details of study characteristics. Overall, 1292 individuals with long-covid were included in the studies (698 in the intervention groups, 594 in the control groups) with a mean age and body-mass index of 49 years and 28 respectively, and a mean treatment duration of 6 weeks. Analysed individuals belonged to all waves of COVID.

Interventions (detailed in the supplementary materials), however numerous, were divided into four distinctive categories (i) High intensity rehabilitation (ii) Low to moderate intensity rehabilitation (iii) Respiratory muscles training and (iv) Others. In the first category, six trials [39,41–45] included an outpatient PR (using cycle ergometer or treadmills) at high intensity for long sessions. Six studies in the second group [46–51] used mixed numerous rehabilitation techniques including but not limited to: aerobic training and strength training at low to moderate intensity, walking, relaxation, stretching and breathing exercises. All of these therapies were home-conducted by videoconference. The third group comprised six studies [52–57] which focused mostly on respiratory muscles specific strength training using thresholds devices and breathing exercises with or without using incentive spirometers. Finally, one study [58] evaluated an alternative treatment modality by Low-Level Laser Therapy (LLLT). The average frequency of treatment was 22 sessions over 6 weeks with a single session duration ranging between 30 min and more than 2 h.

Regarding control interventions, two studies [54,58] used sham interventions, five studies [43,48,51,52,56] used another form of rehabilitation while nine others [41,42,45–47,49,53,55,57] used a “no-rehabilitation” control group and one study [50] used a brochure of self rehabilitation. Finally, two studies [39,44] used pulmonary rehabilitation (PR) delivered in an alternative manner (through virtual reality and through videoconference).

Most studies reported dyspnoea on more than one scale; nine used the mMRC scale, and nine studies used the Borg dyspnoea scale. Other scales, multidimensional, were used such as the Multidimensional Dyspnoea Profile (MDP) and Dyspnoea-12 (D12), the Dyspnoea Severity Index (DSI) or the Pulmonary Functional Status and Dyspnea Questionnaire-Modified (PFSDQ-M).

The PEDro score was ranked “good” for 9 studies, “fair” for 9 studies

Table 1
Characteristics of included studies.

Group	Characteristics							Intervention				
	Study/country	Type	Number of subjects	Female sex (%)	Mean Age (years)	Mean BMI	Inclusion	Duration (weeks)	Number of sessions	Outcome	Intervention group	Control group
Others	Alghitany, 2023 (Egypt)	RCT	80	N/A	36	32	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	12	36	mMRC	Low-Level Laser Therapy	Sham technique
Low-to-moderate intensity	Arora, 2022 (India)	RCT	32	N/A	42	25	Discharged for Covid with residual Dyspnoea	4	14	Borg	Mixed Telerehabiliation at moderate intensity	No rehabilitation
	Fares, 2023 (Egypt)	RCT	100	31	58	31	Discharged for Covid with residual Dyspnoea	6	36	mMRC	Mixed Telerehabilitation at Moderate Intensity	No rehabilitation
	Hajibashi, 2023 (Iran)	RCT	52	46	44	26	Discharged for Covid with residual Dyspnoea	6	30	Borg	Mixed Telerehabiliation at moderate intensity + Progressive Muscle relaxation	Mixed Telerehabiliation at moderate intensity
	Li, 2022 (China)	RCT	120	56	51	N/A	Discharged for Covid with residual Dyspnoea	6	18–24	mMRC	Mixed Telerehabiliation at moderate intensity	No rehabilitation
	Pehlivan, 2022 (Turquey)	RCT	34	26	47	27	Discharged for Covid with residual Dyspnoea	6	18	mMRC	Mixed Telerehabiliation at moderate intensity	Brochure of self rehabilitation
	Şahın, 2023 (Turquey)	RCT	42	25	60	29	Discharged for Covid with residual Dyspnoea	8	30–42	mMRC	Mixed Telerehabiliation at moderate intensity + Coaching phone call once a week	Mixed Telerehabiliation at moderate intensity
High-intensity	Asimakos, 2023 (Greece)	Non-randomised controlled Trial	44	53	56	N/A	Discharged for Covid with residual Dyspnoea	8	16	CAT	Pulmonary rehabilitation	No rehabilitation
	Campos, 2022 (Brazil)	Non-randomised controlled Trial	37	48	43	27	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	8	16	PFSDQ-M	Pulmonary rehabilitation	No rehabilitation
	Jimeno-Almazan, 2023 (Spain)	RCT	83	69	45	26	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	8	16	mMRC	Pulmonary rehabilitation	No rehabilitation
	Romanet, 2023 (France)	RCT	60	37	58	30	Discharged for Covid with residual Dyspnoea	10	20	MDP	Pulmonary rehabilitation	Physiotherapy
	Rutkowski, 2023 (Poland)	RCT	32	63	58	N/A	Discharged for Covid with residual Dyspnoea	3	15	Borg	Pulmonary rehabilitation + VR	Pulmonary rehabilitation
	Vallier, 2023 (France)	RCT	17	29	55	30	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	4	16	mMRC	Home-based Pulmonary rehabilitation	Pulmonary rehabilitation
Respiratory muscles training	Abodonya, 2021 (Egypt)	Non-randomised controlled Trial	42	22	48	28	Discharged for Covid with residual Dyspnoea	2	28	DSI	IMT + incentive spirometry	Incentive spirometry
	De La Plaza San Frutos, 2023 (Spain)	Non-randomised controlled Trial	100	69	49	N/A	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	3	10	MDI	Respiratory Telerehabilitation	Waiting List
	Del Corral, 2023 (Spain)	RCT	88	72	46	N/A	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	8	48	Dyspnoea Y/N	IMT	IMT Sham

(*continued on next page*)

Table 1 (*continued*)

Group	Characteristics							Intervention				
	Study/country	Type	Number of subjects	Female sex (%)	Mean Age (years)	Mean BMI	Inclusion	Duration (weeks)	Number of sessions	Outcome	Intervention group	Control group
	Mc Narry, 2022 (United Kingdom)	RCT	148	88	46	28	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	8	24	TDI	IMT	Waiting List
	Nagy, 2022 (Egypt)	RCT	52	0	40	27	Discharged for Covid with residual Dyspnoea	6	18	mMRC	IMT + Diaphragm Release	IMT
	Philip, 2022 (United Kingdom)	RCT	129	81	49	27	Long-Covid (Hospitalised and non-hospitalised) with residual Dyspnoea	6	6	Borg	Singing	No rehabilitation

Abbreviations: RCT: randomised controlled trial; mMRC: modified medical research council; CAT: chronic obstructive pulmonary disease assessment test; PFSDQ-M: modified pulmonary functional status and *dyspnea* questionnaire; MDP: multidimensional dyspnea profile; DSI: dyspnea severity index; MDI: Mahler's dyspnea index; TDI: transition dyspnea index; IMT: inspiratory muscle training; VR: virtual reality.

and "bad" for one study (Fig. 3). As expected, blinding of participants and therapists was seldom achieved in these trials. The funnel plot (Fig. 4) shows a homogeneous literature distribution, consistent with an overall absence of publication bias.

Random-effects meta-analysis (n = 17) showed a significant positive treatment effect on dyspnoea post-intervention in favour of physiotherapy (SMD -0.63, 95 CI [−1.03; −0.24], $p < 0.001$, $I^2 = 88$ %) despite a significant heterogeneity (Fig. 5). This effect was only confirmed in the High intensity rehabilitation group during sub-group analysis which showed null heterogeneity ($I^2 = 0$ %). As only one study evaluated a passive technique, no meta-analysis could be made on this subgroup. Results show a significant effect in favour of intervention compared to a sham technique.

4. Discussion

This systematic review of 19 comparative trials provides evidence favouring physiotherapy to decrease persisting dyspnoea following COVID-19. Furthermore, while numerous interventions were used, this effect is only found in the sub-group "high intensity rehabilitation".

Results of the low-to-moderate intensity and RMT groups studies are non-significant, despite a fair number of people included and a favourable trend towards rehabilitation. A wide combination of various techniques were used by the authors, some issued from physical therapy (e.g. endurance and exercise training), others from respiratory therapy (e.g. active cycle breathing technique) and then some based on the psycho-affective aspect of dyspnoea (e.g. deep breathing relaxation), making the interpretation of each study independently impractical. These studies are in fact highly heterogeneous, not only in terms of techniques used, but also in the duration of each session, the presence or absence of a therapist as well as the intensity of effort involved, ranging from no effort (relaxation) to moderate intensity (brisk walking or moderate-load resistance training). Treatment heterogeneity has been and still is a well known factor in physiotherapy [25]. While this is a major limitation for researchers carrying out systematic reviews and meta-analyses, from the physiotherapist's point of view these numerous treatment modalities represent a formidable therapeutic arsenal on all clinical manifestations of disability of one individual, allowing a "tailor-made" treatment to be found [26,27]. It thus appears likely that if an effect could be present on the individual scale over long periods of care (as rehabilitation is engineered over several months), its size effect would be too little, even in a population as homogeneous as those with long-covid. Furthermore, the large number of methods used may help define common interventions that could form the basis of future standardised interventions.

Interestingly, the only effect was found in the high-intensity group which shared similar interventions with homogeneity. Rehabilitation was systematically modelled on PR, associating therapeutic education, self rehabilitation and long sessions (over 90 min) of both endurance and exercise training at high intensity at least twice a week. PR, the key component of which is high intensity training, is defined by the American Thoracic Society/European Respiratory Society (ATS/ERS) statement and is supported by a level 1A evidence for COPD to reduce dyspnoea, improve exercise capacity, health related quality of life and reduce hospital admissions [22,59]. Since its effect is based on effort and proportional to it, finding the same effect in long-covid is not only expected but also supports the idea that PR probably performs in the same way in both COPD and long-covid. Furthermore the fact that it is correlated with intensity tends to confirm that one of the key components of dyspnoea in long-covid is a functional component (deconditioning), in addition to other mechanisms. Indeed, dyspnoea, like pain, is complex and multifactorial, as extensively studied in COPD [60]. Observational studies show that diffusion capacity (diffusing capacity for carbon monoxide) and pulmonary function (forced expiratory volume, vital capacity) can remain impaired months after COVID, regardless of initial severity [49,61–63]. The effects of PR are however not so

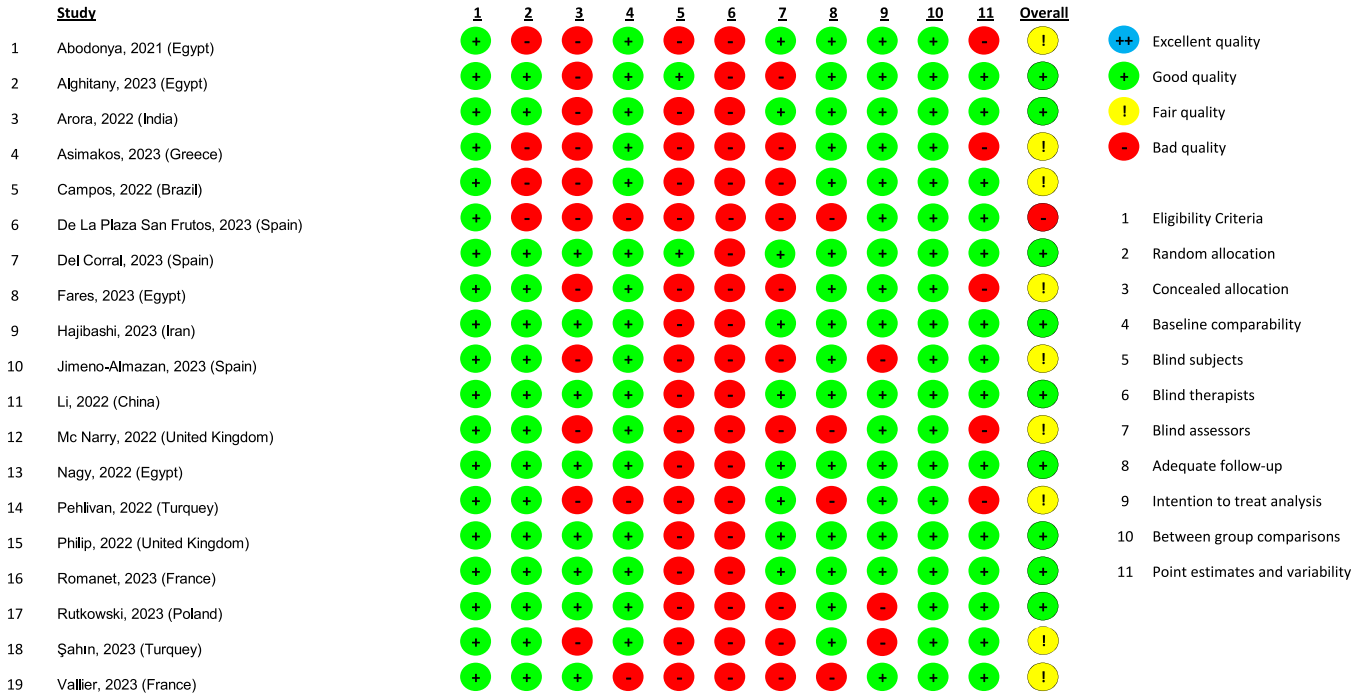

Fig. 3. Risk of bias assessment, following the PEDRO scale.

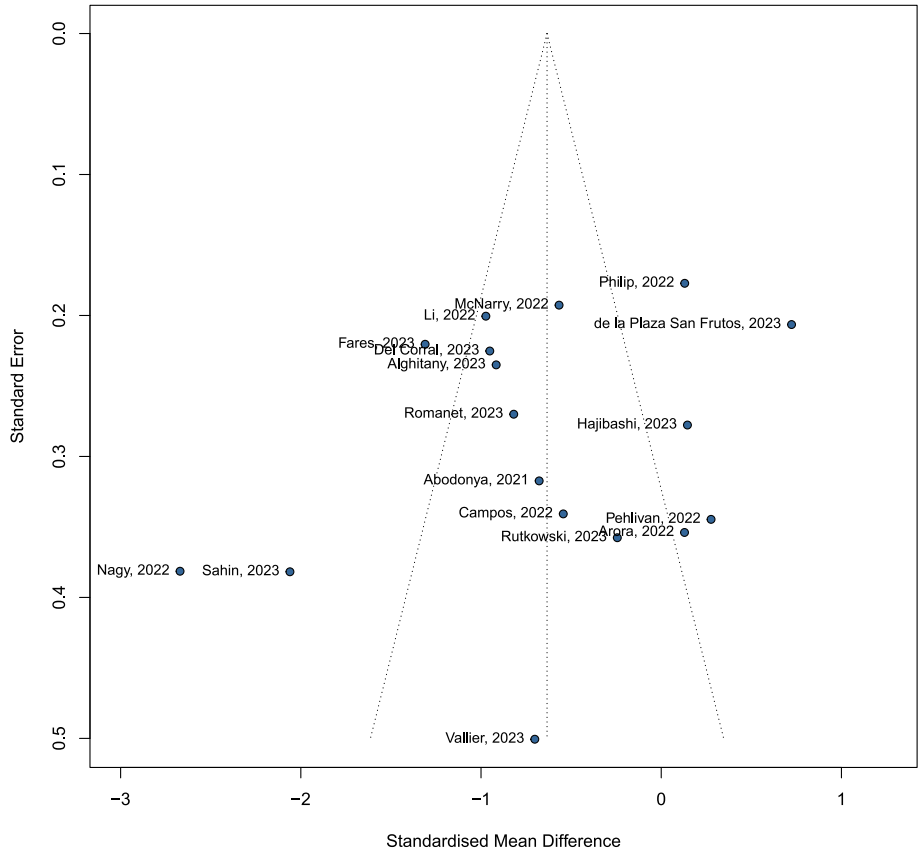

Fig. 4. Funnel plot assessment for publication bias.

quite clear on those later outcomes in COPD or interstitial diseases [64], as alveolar damage is generally irreversible, and too few studies have thus explored rehabilitation's impact on these aspects in long COVID [65,66].

Unfortunately, while PR remains a major course of therapy for people with lung diseases, its actual availability for individuals is sub-optimal worldwide: less than 3 % of people with chronic respiratory diseases actually have such access [67–70]. Too few practices have the

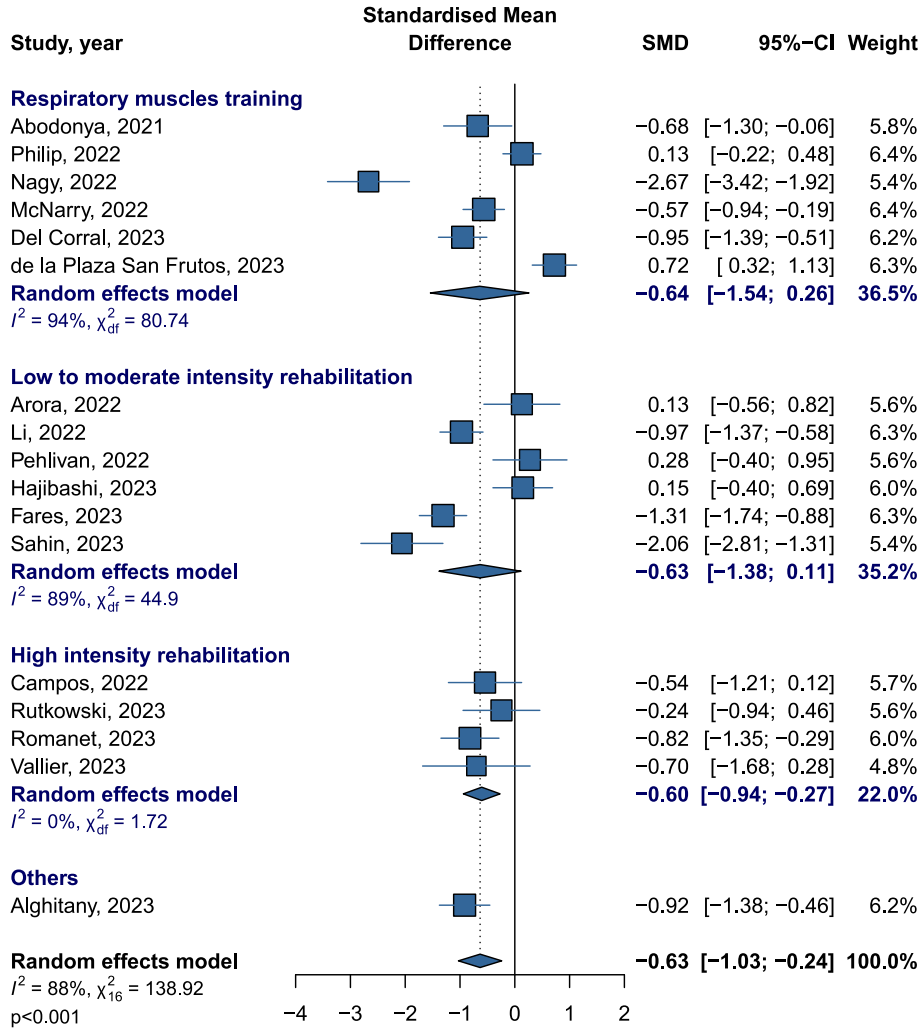

Fig. 5. Random effect subgroup meta-analysis of comparative trials examining the effect of physiotherapy interventions versus control; a negative value favours intervention for reducing dyspnoea.

training, equipment and space to properly conduct PR and many physicians are still not aware of this rehabilitation modality. Several authors have put forward the scarcity of PR practices in Australia [71], Sweden [68] or in the USA [72,73] which counts 1 PR practice for almost 200 000 inhabitants. For example, in and around Paris, France in 2022, only 106 PR practices were registered for a population of over 12 million people and people often had to be put on waitlists, with 20 % of them eventually opting out of care [74]. While the COVID-19 pandemia has highlighted this discrepancy, efforts are still to be made to properly respond to the actual demand [43,59].

It is worth noticing that two studies [39,44] compared PR with two alternate modalities of PR (PR coupled with virtual reality and home-based tele-PR) and showed a significant effect in favour of the alternate modality. While those results, if unexpected, should be pondered by the open design of both trials, they could be explained by a better adhesion to treatment or by the additional affective dimension of dyspnoea. This might suggest that PR carried out in a more welcoming environment or by diverting attention away from the efforts being made may be more effective than PR alone in an outpatient setting. Future studies may shed light on this peculiar setting [75].

In addition, home-based rehabilitation has increasingly been put forward as a remedy for the issues of healthcare availability [76]. It provides people with guaranteed access and made-to-measure planning, thereby improving compliance while simultaneously preventing the risk of healthcare-related contaminations. Therapeutic groups can still be made, with supervision, or without in the case of autonomous rehabilitation. With many benefits for the public and effects similar to conventional care [77], home-based rehabilitation has logically been tried with success in post-surgery [78,79], COPD [80–82], chronic heart failure [83] and long-covid [44] and most studies in the low-to-moderate intensity and RMT groups have opted for this modality. As recently put forward by the previously cited meta-analyses, home--based rehabilitation seems to be at worst non-significantly different when compared to out or in-patient settings, and may actually be superior thanks to a greater adhesion and availability. However, home-based rehabilitation suffers from the same flaws as outpatient rehabilitation and a substantial heterogeneity caused by too many different settings could have limited the contribution of those sub-groups. Whether using a protocol with too low an intensity may have also prevented a consistent effect, similar to those of Vallier et al., home-based rehabilitation could be considered as a viable measure for those unable to follow rehabilitation [44]. Further studies, with more homogeneous protocols are needed to assess the effectiveness of alternative rehabilitation models, with the "traditional model" remaining the gold standard [84].

As an alternative to whole body rehabilitation the RMT group studies chose to address dyspnoea specifically by treating respiratory muscles but failed to show a significant effect, despite a trend favouring rehabilitation and a large number of subjects recruited. The interventions are characterised by diverse respiratory therapy techniques, inspiratory

(IMT) and/or expiratory (EMT), the modalities of which are quite heterogeneous. At both ends of the spectrum, one study [57] chose to use online group lullaby singing sessions 1 h per week during 6 weeks while another [54] chose a robust double blinded design comparing IMT and EMT to their sham techniques, 40 min per day for 8 weeks. Overall, of the two studies that show the smallest difference compared with the control group, De la Plaza San Frutos et al. have the highest risk of bias of our systematic review and use low intensity RMT for a short duration while Philip et al. used the aforementioned singing, also low in intensity [53,57]. Hence it appears likely that this subgroup once again lacks proper intensity and homogeneity in their treatment. Those results of RMT are consistent with those found on COPD which is to this day unable to show a significant effect on dyspnoea, quality of life, physical fitness and muscle strength, particularly when administered alongside PR, presumably on account of the greater effectiveness of the latter [85]. Indeed, the protocols of the RMT group all compared RMT with either no treatment, or RMT itself, allowing for a small effect to be found favouring intervention, as it is described [85]. Yet, from a biomechanical perspective, respiratory muscle dysfunction has been suggested as a potential contributing factor to chronic breathlessness following COVID-19, as in COPD [9,85]. Studies indeed show that in the case of respiratory diseases, the peak strength of the respiratory muscles may be reduced, either by the disease itself (inflammation, overuse, change of metabolism) or by its treatment (sedentarity, bed rest, mechanical ventilation) [86]. If it makes sense that dyspnoea could be alleviated by stronger, more resilient respiratory muscles thus reducing air hunger and work of breathing, future studies should strive to ensure consistency in the devices used, as well as the intensity, frequency and duration of treatment, in order to achieve a clearer effect.

Finally, only one study evaluated a passive technique [58]. LLLT (also known as laser acupuncture) is a form of phototherapy that has been described mainly as a treatment for musculoskeletal pain with moderate evidence [87,88]. While it does not require the active participation of the person, it seems to have significantly reduced dyspnoea and markers of inflammation in people with long-covid. It has been suggested that LLLT could have an influence on dyspnoea by stimulating the vagus nerve, effectively decreasing the release of acetylcholine in the lungs and thus enabling a bronchodilator and anti-inflammatory effect [89]. Both residual lung and systemic inflammation has strongly been associated with altered pulmonary function tests and the persistence of dyspnoea [6]. If this pathway hypothesis has yet to be fully investigated, it would mean that contrary to the other studies, dyspnoea has been partially dealt with directly through the vagal mechanoreceptors (c-fibres, slow and rapidly adapting receptors) rather than physical reconditioning. Dyspnoea is indeed once again multifactorial [16,17] and no physiotherapy intervention has yet managed to make it disappear through physical techniques alone. If this study has limitations and needs to be accompanied by other RCTs to confirm its results, it nevertheless deserves to be considered as a viable therapy, complementary to physical interventions.

This meta-analysis has some limitations of its own. First, if we acknowledge the near-impossible task of conducting double-blind studies in physiotherapy, the risk of bias arising from open or assessor-only blind studies such as it is the case in all included studies should be taken into consideration as the real effect of interventions might be overestimated, especially considering subjectives outcomes such as dyspnoea. However, this effect is accompanied in most analysed studies by objective outcomes (e.g. 6 min walking test, peak oxygen consumption) following the global same trend. Second, while people with long-covid represent a singular population, characterised by a high proportion of women, of subjects between 30 and 60 years of age with a history of inflammatory diseases as well as overweight subjects thus limiting external validity of findings to other chronic respiratory diseases, the population's homogeneity analysed in our systematic review strengthens the evidence of benefit within the long-covid population [90,91]. Third, intervention and control groups were highly heterogeneous and made it difficult to interpret results. The heterogeneity of rehabilitation settings and methods has long been a significant source of confusion, even among physiotherapists, and continues to hinder the ability to make clear recommendations or draw robust conclusions regarding treatment effects in clinical trials or meta-analyses. Another complicating factor is the often insufficient reporting of key details related to the intervention, control group, or patient population in these trials, which limits the ability to carry out comprehensive analyses of what was really implemented and to whom it benefited the most. Despite universal global recommendations, highly diverse methods have been employed in rehabilitation interventions, a trend that was quite clear in our systematic review. While the interventions we reviewed were heterogeneous, we observed that sorting studies in subgroups following the intensity of efforts and the type of rehabilitation carried has allowed us to compare one rehabilitation component with another, in particular the effects of different operating methods. Consequently, it seems that intensity and protocolization appear to be key factors contributing to effectiveness. Specifically, studies that adhered to high-intensity rehabilitation protocols in line with COPD guidelines for pulmonary rehabilitation demonstrated both significant and homogeneous improvements in dyspnoea. Although the mechanisms underlying dyspnoea are complex and likely differ between COPD and COVID-19, there may be shared underlying factors that warrant further exploration. Last, even if the intervention periods are modelled on actual practice and the analyses were often made in intention-to-treat thus representing rehabilitation as carried out in real life, our findings only focused on the short term and do not represent the evolution of dyspnoea months or years after rehabilitation.

5. Conclusion

In summary, in people with persisting dyspnoea following COVID-19, physiotherapy interventions retrieved in this review showed significant short term improvement in dyspnoea outcomes. Furthermore, high-intensity rehabilitation shows particularly more promising results when compared with low-to-moderate rehabilitation or RMT. Future studies on physiotherapy should focus on reducing heterogeneity in both intervention and control groups; use multidimensional dyspnoea models to further investigate which technique would be the most appropriate for a given singular individual and lastly, strive to describe precisely the intervention actually administered to the subjects.

CRediT authorship contribution statement

Christophe Romanet: Writing – original draft, Supervision, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Johan Wormser:** Writing – review & editing, Investigation, Data curation. **Marine Cachanado:** Visualization, Software, Methodology, Formal analysis, Data curation. **María Granados Santiago:** Writing – review & editing, Visualization, Supervision. **Gilles Chatellier:** Writing – review & editing, Validation, Software, Methodology, Formal analysis, Data curation. **Marie Carmen Valenza:** Writing – review & editing, Visualization, Supervision. **François Philippart:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CR, JW and FP contributed to study conception, literature search, and data extraction; CR, JW, FP and GC contributed to assessment of risk of bias; MC and GC contributed to the statistical analysis and methodology; CR drafted the manuscript, and JW, FP, MGS and MCV contributed to critical review.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rmed.2024.107909.

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